

**SECONDARY 1 FINAL EXAMINATION**  
**MATHEMATICS COMPULSORY PART**  
**SUGGESTED SOLUTIONS AND MARKING SCHEME**

**SECTION A: MULTIPLE CHOICE SOLUTIONS (20 Marks)**

| Q. No. | Option | Full Solution & Explanation                                                                                                                   |
|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | B      | $\text{L.C.M.} = 2^{\max(2,1,0)} \times 3^{\max(1,2,1)} \times 5^{\max(1,0,2)} \times 7^{\max(0,1,0)}$ $= 2^2 \times 3^2 \times 5^2 \times 7$ |
| 2      | B      | $1.8 \div \left(\frac{2}{5} + \frac{1}{2}\right) - 0.5 = 1.8 \div (0.4 + 0.5) - 0.5$ $= 1.8 \div 0.9 - 0.5$ $= 2 - 0.5$ $= 1.5$               |
| 3      | A      | $(-2)^3 - (-5) \times (-3) + (-12) \div 4 = -8 - 15 + (-3)$ $= -26$                                                                           |
| 4      | A      | $\frac{x^2 - 3y}{x - y} = \frac{(-3)^2 - 3(-2)}{-3 - (-2)}$ $= \frac{9 + 6}{-1}$ $= -15$                                                      |
| 5      | A      | $(3a^2b^{-1})^2 \times (2ab^3)^{-1} = \frac{9a^4b^{-2}}{2ab^3}$ $= \frac{9a^3}{2b^5}$                                                         |
| 6      | B      | $\frac{4^{x+1} \times 2^x}{8^x} = \frac{2^{2(x+1)} \times 2^x}{2^{3x}}$ $= \frac{2^{3x+2}}{2^{3x}}$ $= 2^2$ $= 4$                             |
| 7      | C      | $\frac{3x-1}{4} - \frac{1}{3} = \frac{5}{3}$ $\frac{3x-1}{4} = 2$ $3x-1 = 8$ $x = 3$                                                          |
| 8      | B      | Degree of term $5x^3y^2$ is $3 + 2 = 5$ .<br>Total number of terms in $5x^3y^2 - 3x^2y + 8x - 12$ is 4.                                       |
| 9      | A      | Arranging $4x - 2 + 7x^3 - x^2$ in ascending powers of $x$ : $-2 + 4x - x^2 + 7x^3$ .                                                         |
| 10     | B      | $200 \times (1 + 15\%) + 150 \times (1 - 20\%) = 200(1.15) + 150(0.80)$ $= 230 + 120$ $= 350$                                                 |

| Q. No. | Option | Full Solution & Explanation                                                                                                                                                                                                    |
|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11     | A      | $\begin{aligned} \text{Overall percentage factor} &= (1 + 20\%)(1 - 15\%) \\ &= 1.20 \times 0.85 \\ &= 1.02 \\ &= 1 + 2\% \end{aligned}$                                                                                       |
| 12     | C      | $\begin{aligned} \text{Percentage of boys} &= 100\% - 65\% \\ &= 35\% \\ \text{Total members} &= \frac{28}{35\%} \\ &= 80 \end{aligned}$                                                                                       |
| 13     | D      | $\begin{aligned} \text{Sum of interior angles of a hexagon} &= (6 - 2) \times 180^\circ \\ &= 720^\circ \end{aligned}$                                                                                                         |
| 14     | B      | $\begin{aligned} \text{Mean} &= \frac{52 + 58 + 61 + 64 + 67 + 70 + 73 + 75 + 82 + 88}{10} \\ &= \frac{690}{10} \\ &= 69.0 \end{aligned}$                                                                                      |
| 15     | B      | $\begin{aligned} \text{Arc length} &= 2\pi(9) \times \frac{80^\circ}{360^\circ} \\ &= 4\pi \text{ cm} \end{aligned}$                                                                                                           |
| 16     | C      | $\begin{aligned} \text{Area} &= \pi r^2 \times \frac{\theta}{360^\circ} \\ 15\pi &= \pi(6)^2 \times \frac{\theta}{360^\circ} \\ \theta &= 150^\circ \end{aligned}$                                                             |
| 17     | D      | $\begin{aligned} \text{In } \triangle ABC, \angle ACB &= 180^\circ - 45^\circ - 65^\circ = 70^\circ. \\ \text{Since } AB \parallel CD, \angle ACD &= 180^\circ - \angle BAC = 180^\circ - 45^\circ = 135^\circ. \end{aligned}$ |
| 18     | A      | $\begin{aligned} \text{Twice } x \text{ minus 4 is } &2x - 4. \\ \text{Dividing by sum of } y \text{ and 3 gives } &\frac{2x-4}{y+3}. \end{aligned}$                                                                           |
| 19     | B      | $\begin{aligned} \text{Total cost} &= 3x + 5y. \\ \text{Change} &= 200 - (3x + 5y) = \$(200 - 3x - 5y). \end{aligned}$                                                                                                         |
| 20     | A      | $\begin{aligned} \text{Base area} &= \frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2. \\ \text{Volume} &= 12 \times 10 = 120 \text{ cm}^3. \end{aligned}$                                                                      |

## SECTION B: COMPULSORY LONG QUESTIONS (40 Marks)

| Q. No.    | Solution Steps                                                                                                                                               | Marks    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 21(a)     | $\frac{3}{4} + \frac{5}{6} \times \frac{3}{10} = \frac{3}{4} + \frac{1}{4}$ $= 1$                                                                            | 1M<br>1A |
| 21(b)     | $2.4 \div \left(1\frac{1}{2} - 0.3\right) = 2.4 \div (1.5 - 0.3)$ $= 2.4 \div 1.2$ $= 2$                                                                     | 1M<br>1A |
| 21(c)     | $\frac{1}{1 + \frac{1}{1+\frac{1}{2}}} = \frac{1}{1 + \frac{1}{3/2}}$ $= \frac{1}{1 + \frac{2}{3}}$ $= \frac{1}{5/3}$ $= \frac{3}{5} \quad (\text{or } 0.6)$ | 1M<br>1A |
| 22(a)     | Temperature rise = $+5^{\circ}\text{C}$                                                                                                                      | 1A       |
|           | Temperature drop = $-9^{\circ}\text{C}$                                                                                                                      | 1A       |
| 22(b)     | Final temperature = $-12 + 5 - 9$                                                                                                                            | 1M       |
|           | $= -16^{\circ}\text{C}$                                                                                                                                      | 1A       |
| 22(c)     | Required change = $-30 - (-16)$                                                                                                                              | 1M       |
|           | $= -14^{\circ}\text{C} \quad (\text{or a drop of } 14^{\circ}\text{C})$                                                                                      | 1A       |
| 23(a)     | $(x^3y^{-2})^3 = x^{3 \times 3}y^{-2 \times 3}$ $= x^9y^{-6}$ $= \frac{x^9}{y^6}$                                                                            | 1M<br>1A |
| 23(b)     | $\frac{18a^4b^2}{6a^{-2}b^5} = 3a^{4-(-2)}b^{2-5}$ $= 3a^6b^{-3}$ $= \frac{3a^6}{b^3}$                                                                       | 1M<br>1A |
| 23(c)     | $\left(\frac{2m^2}{n^{-1}}\right)^3 \div (4m^4n^2) = \frac{8m^6n^3}{4m^4n^2}$ $= 2m^{6-4}n^{3-2}$ $= 2m^2n$                                                  | 1M<br>1A |
| 24(a)(i)  | $3(2x - 5) = 4x + 1$ $6x - 15 = 4x + 1$ $2x = 16$ $x = 8$                                                                                                    | 1M<br>1A |
| 24(a)(ii) | $\frac{x+3}{2} - \frac{2x-1}{5} = 2$ $5(x+3) - 2(2x-1) = 20$ $5x + 15 - 4x + 2 = 20$ $x + 17 = 20$ $x = 3$                                                   | 1M<br>1A |

| Q. No.    | Solution Steps                                                                                                                                                                                           | Marks                      |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 24(b)     | $(2x - 3)(x^2 + 3x - 2) = 2x(x^2 + 3x - 2) - 3(x^2 + 3x - 2)$ $= 2x^3 + 6x^2 - 4x - 3x^2 - 9x + 6$ $= 2x^3 + 3x^2 - 13x + 6$ <p>The coefficient of <math>x^2 = 3</math></p>                              | 1M<br><br><br>1A<br><br>1A |
| 25(a)     | <p>Marked price = <math>500 \times (1 + 40\%)</math></p> <p>= \$700</p>                                                                                                                                  | 1M<br>1A                   |
| 25(b)(i)  | <p>Selling price = <math>700 \times (1 - 20\%)</math></p> <p>= \$560</p>                                                                                                                                 | 1M<br>1A                   |
| 25(b)(ii) | <p>Profit percentage = <math>\frac{560 - 500}{500} \times 100\%</math></p> <p>= 12%</p>                                                                                                                  | 1M<br>1A                   |
| 26(a)     | <p>Arc length <math>AB = 2\pi(10) \times \frac{108^\circ}{360^\circ}</math></p> <p>= <math>6\pi</math> cm</p>                                                                                            | 1M<br>1A                   |
| 26(b)     | <p>Perimeter = <math>6\pi + 10 + 10</math></p> <p><math>\approx 38.8</math> cm (to 3 sig. fig.)</p>                                                                                                      | 1M<br>1A                   |
| 27(a)     | <p>Number of students with mark <math>\geq 70</math> in Class 1A = 8</p> <p>Percentage = <math>\frac{8}{20} \times 100\%</math></p> <p>= 40%</p>                                                         | 1M<br><br>1A               |
| 27(b)     | <p>Proportion of Distinction in Class 1B = <math>\frac{126^\circ}{360^\circ} \times 100\%</math></p> <p>= 35%</p> <p><math>\therefore 40\% &gt; 35\%</math></p> <p><math>\therefore</math> Class 1A.</p> | 1M<br>1A<br>1M<br>1A       |

## SECTION C: CHALLENGING QUESTIONS (8 Marks)

| Q. No.    | Solution Steps                                                                                                      | Marks |
|-----------|---------------------------------------------------------------------------------------------------------------------|-------|
| 28(a)     | $\angle AEF + \angle CFE = 180^\circ$ (Given)                                                                       | 1M    |
|           | $\therefore AB \parallel CD$ (int. $\angle$ s supp.)                                                                | 2A    |
| 28(b)(i)  | $\angle EFD = 180^\circ - \angle CFE = \angle AEF = 110^\circ$ (alt. $\angle$ s, $AB \parallel CD$ )                | 1M    |
|           | $\angle EFG = 180^\circ - 110^\circ = 70^\circ$ (adj. $\angle$ s on st. line)                                       | 1M    |
|           | In $\triangle EFG$ , $\angle EGF = 180^\circ - 40^\circ - 70^\circ$ ( $\angle$ sum of $\triangle$ )<br>$= 70^\circ$ | 1A    |
| 28(b)(ii) | Since $\angle EFG = \angle EGF = 70^\circ$ ,                                                                        | 1M    |
|           | $EF = EG$ (sides opp. equal $\angle$ s).<br>$\therefore \triangle EFG$ is an isosceles triangle.                    | 1A    |
| 29(a)     | Base area = $\frac{(AD + BC) \times AB}{2}$                                                                         | 1M    |
|           | $= \frac{(x + x + 4) \times 6}{2}$                                                                                  | 1A    |
|           | $= (6x + 12) \text{ cm}^2$                                                                                          |       |
| 29(b)     | Volume = $(6x + 12) \times 12 = 432$                                                                                | 1M    |
|           | $72x + 144 = 432$                                                                                                   | 1M    |
|           | $72x = 288$                                                                                                         |       |
|           | $x = 4$                                                                                                             | 1A    |
| 29(c)     | When $x = 4$ , $AD = 4$ cm.                                                                                         |       |
|           | Base dimensions of max cuboid = $AB \times AD$                                                                      | 1M    |
|           | $= 6 \text{ cm} \times 4 \text{ cm}$                                                                                |       |
|           | Maximum volume = $6 \times 4 \times 12$                                                                             | 1M    |
| 30(a)     | $2520 = 2^3 \times 3^2 \times 5 \times 7$                                                                           | 2M    |
|           | $k = 2 \times 5 \times 7$                                                                                           | 1M    |
|           | $= 70$                                                                                                              | 1A    |
|           |                                                                                                                     |       |
| 30(b)     | $r^2 = \frac{2520k}{49}$                                                                                            | 1M    |
|           | $= \frac{2520 \times 70}{49}$                                                                                       |       |
|           | $= 3600$                                                                                                            | 1M    |
|           | $x = \pm 60$                                                                                                        |       |
|           |                                                                                                                     | 2A    |

– End of Solution –